

Highlights

DeXposure-FM: A Time-series, Graph Foundation Model on Credit Exposures and Stability on Decentralised Financial Networks

- Financial economics foundation model: We build DeXposure-FM, the first time-series, graph, tabular foundation model in DeFi and the wider financial economic space.
- Financial economics / ML benchmark: We show that DeXposure-FM produces competitive multi-step forecasts of edge-level exposures and network-level statistics (concentration, density, sector-to-sector links) relative to econometric and machine learning baselines (factor-VARs, temporal GNNs).
- Economic measurement and systemic-risk statistics for DeFi: We use DeXposure-FM to derive richer systemic-risk measures—protocol-level systemic-importance scores, sectoral spillover indices, and early-warning indicators based on shifts in network structure and reliance on fragile collateral-providing information beyond standard size- and TVL-based metrics and supporting macroprudential surveillance and DeFi regulation.
- Financial risk stress testing: Using the model, we generate shock-consistent scenarios for major DeFi stress events (e.g. stablecoin de-pegs, exchange failures) and quantify loss distributions, contagion paths, and the impact of counterfactual policy tools (leverage caps, collateral haircuts, bridge limits).

DeXposure-FM: A Time-series, Graph Foundation Model on Credit Exposures and Stability on Decentralised Financial Networks

Abstract

We introduce DeXposure-FM, a time-series, graph, tabular foundation model for measuring and forecasting inter-protocol credit exposures in global decentralised financial networks. The model has an architecture of xxx. The model is built over a large-scale dataset DeXposure, which reconstructs value-linked credit exposures (measured by Total Value Locked, TVL) and token flows across 4,300+ protocols on 602 blockchains, covering 24,300+ unique tokens over 2020–2025. DeXposure-FM is trained for the task of edge-level credit-exposure forecasting by an advanced optimiser, Sharpness-Aware Minimization (SAM), predict the joint dynamics of (i) protocol-level TVL and flows, (ii) the topology and weights of credit-exposure links, and (iii) sector structure (lending, exchanges, asset management, bridges, stablecoins, etc.).

We evaluate DeXposure-FM on a suite of machine-learning benchmarks: (1) multi-step forecasting of edge-level exposures and network-level statistics, including concentration, density, and sector connectivity; (2) policy-relevant scenario analysis, where the model generates shock-consistent paths of losses and contagion during major stress events; and (3) imputation and denoising of missing or noisy exposures and TVL histories, with TVL-weighted error metrics and downstream performance on forecasting and link-prediction tasks. We compare DeXposure-FM to baselines including VAR models, low-rank factor models, and temporal graph neural networks.

On the financial economics front, we use the world model to construct dynamic measures of (1) protocol-level systemic importance, (2) sector-to-sector spillover indices, and (3) early-warning indicators based on shifts in network concentration and dependence on fragile collateral. We show how these AI-generated statistics can support macroprudential monitoring, DeFi-specific stress testing, and counterfactual policy evaluation (e.g., caps on bridges or leverage and alternative collateral rules), while also highlighting

conditions under which foundation-model-based measures of digital financial infrastructure are robust—and where they call for cautious interpretation by regulators and central banks.

Model:

Code: <https://github.com/EVIEHub/DeXposure-FM>

Keywords: Time-Series Analysis, Graph Tabular Model, Foundation Model, Credit Exposures, Financial Stability, Decentralised Finance, Financial Networks

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1. Introduction

Decentralised finance (DeFi) has emerged as a sizeable and fast-evolving segment of the crypto-asset ecosystem, offering lending, trading, derivatives and payment-like services via smart contracts deployed across multiple blockchains [1, 2]. A defining feature of DeFi is composability: protocols routinely hold, lock, or accept tokens issued by other protocols as collateral, liquidity, or reserves. As a result, credit exposure is often implicit and token-mediated rather than expressed as bilateral contracts, creating a dense web of inter-protocol dependencies [3, 4]. When a token that is widely used as collateral or liquidity experiences a shock—through a price collapse, governance failure, exploit, or liquidation event—losses can propagate to protocols that rely on that token, potentially generating amplification and contagion across chains [5, 2]. Despite the policy relevance of these network effects, empirical work has been constrained by limited coverage and static snapshots, leaving a gap for dynamic, ecosystem-wide measurement that can track how exposure concentrations and transmission channels evolve over time [3].

Policy bodies have repeatedly highlighted that DeFi can replicate traditional financial functions while inheriting familiar vulnerabilities, leverage, liquidity and maturity transformation, operational fragilities, and interconnectedness, often in forms that can unwind abruptly because key risk-management processes are automated in code [6, 7]. In particular, collateralised lending protocols rely on oracles and liquidation mechanisms that can trigger rapid deleveraging and fire-sale dynamics when prices move sharply or data feeds are disrupted [8, 9, 10]. Stablecoins are a central node in these dynamics: they serve as settlement assets and collateral across protocols, and their de-pegs and runs can transmit stress through both on-chain and off-chain channels [11, 12, 13]. Yet regulators and researchers emphasise persistent data gaps that hinder monitoring and the construction of standardised risk indicators for the DeFi ecosystem [6, 7]. This measurement gap is increasingly salient as links between crypto markets and the broader financial system deepen [13].

To address this gap, we train **DeXposure-FM**, a time-series, graph, and tabular foundation model for measuring and forecasting inter-protocol credit

exposures in global decentralized financial networks. The model is trained on the **DeXposure** dataset, which reconstructs value-linked credit exposures and token flows across more than 4,300 protocols on 602 blockchains, covering over 24,300 tokens from 2020–2025 [3]. DeXposure operationalizes inter-protocol exposures by linking protocol balance-sheet dynamics (e.g., TVL) to token- and sector-level structure, enabling a unified view of volumes, topology, and compositional risk. DeXposure-FM follows the foundation-model paradigm: it pretrains a general representation on heterogeneous data and reuses it across downstream tasks, a strategy that has proven effective in time-series and structured-data settings [14, 15]. We train the model with Sharpness-Aware Minimization (SAM) to improve out-of-sample robustness by favoring parameter regions with uniformly low loss [16].

We evaluate DeXposure-FM on a suite of machine-learning benchmarks tailored to economic measurement and risk analytics. First, we perform multi-step forecasting of edge-level exposures and network-level statistics (e.g., concentration, density, and sector connectivity). Second, we conduct policy-relevant scenario analysis by generating shock-consistent paths of losses and contagion around major stress events (e.g., stablecoin de-pegs and exchange failures). Third, we study imputation and denoising of missing or noisy exposure and TVL histories using TVL-weighted metrics and downstream performance on forecasting and link-prediction tasks. Across tasks, we benchmark against classical econometric and latent-structure baselines (e.g., VAR and factor models) as well as temporal graph neural networks.

On the financial economics front, DeXposure-FM provides a generative measurement layer for macroprudential monitoring in digital financial infrastructure. We use the trained model to construct dynamic measures of protocol-level systemic importance, sector-to-sector spillover indices, and early-warning indicators based on shifts in network concentration and dependence on fragile collateral. These AI-generated statistics support DeFi-specific stress testing and counterfactual policy evaluation (e.g., caps on bridge exposures or leverage, and alternative collateral rules), while clarifying when model-based measures are robust and when they require cautious interpretation by regulators and central banks. By combining novel on-chain data with a multimodal foundation model, DeXposure-FM advances the measurement toolkit needed to understand systemic risk in DeFi and its interaction with the wider financial system [17, 18].

2. Related Works

2.1. World Models and Foundation Models for Economic Forecasting

Recent advances in AI have seen the rise of *world models*—predictive models that capture the dynamics of complex systems. Such world models aim to simulate or forecast the state of the environment, effectively allowing an AI to “understand” or predict future outcomes [19]. In economics, similar ideas have emerged under the framework of large pre-trained forecasting models. [20] discuss the latest generation of time-series large language models (TSLMs) or time-series foundation models (TSFMs), which are large transformer-based models pretrained on diverse financial and economic data. Examples include Google’s TimesFM [21], Salesforce’s Moirai (Woo et al., 2024), and IBM’s Tiny Time Mixers, all demonstrating that such models can outperform traditional forecasting baselines. Notably, [22] introduce *FinCast*, the first foundation model specifically designed for financial time-series, showing robust zero-shot forecasting performance and substantially improving on previous state-of-the-art methods. These foundation models leverage vast pretraining data and can often be prompted or fine-tuned with a few examples (in-context learning) to adapt to new economic forecasting tasks [23].

2.2. Credit Risk Modeling in Traditional and Decentralized Finance

In traditional finance, credit risk modeling spans both individual-default models and systemic network approaches. Classical structural models (e.g., Merton) and reduced-form models quantify default probabilities of single entities, while network contagion models examine how defaults propagate through financial institutions. In the decentralized finance domain, credit risk manifests differently, as loans are typically over-collateralized and mediated by smart contracts. [4] propose simple aggregate risk measures for DeFi lending protocols, using only total deposits and borrowings, and report that systemic fragility surged around mid-2021 during crypto market turmoil. Complementing these measures, [?] compile the *DeXposure* dataset, covering millions of entries across 4.3k protocols. They define inter-protocol credit exposures based on changes in Total Value Locked (TVL), enabling new graph-based analysis of DeFi credit networks. Conceptual studies have begun to map TradFi and DeFi risks together: [5] review systemic risk mechanisms, finding that while basic risk types (leverage, liquidity shocks, corre-

lated exposures) are common to both systems, DeFi’s algorithmic execution and composability lead to unique propagation channels.

2.3. Machine Learning for Systemic Risk, Contagion, and Network Analysis

Machine learning techniques are increasingly applied to systemic risk detection and contagion modeling. Graph neural networks (GNNs) in particular leverage network structure to improve risk prediction. [24] show that a GNN-based model dramatically outperforms traditional ML for classifying systemic importance of banks in simulated networks: GNNs achieved a 94% higher correlation metric on average compared to standard ML classifiers. Likewise, [25] integrate GNNs with explicit interbank liability networks, computing systemic risk measures as the minimum capital needed to secure the system. Their framework effectively learns contagion channels (e.g., Eisenberg–Noe clearing) by incorporating graph-structured data into the risk aggregation function. These modern ML approaches build on network science foundations. Overall, machine-learning-driven network analysis enables end-to-end forecasting of systemic events, aiming to detect contagion pathways and provide early warnings for financial crises.

2.4. Graph and Tabular Foundation Models in Finance/DeFi

Emerging foundation-model architectures for graph and tabular data hold promise for financial applications. Transformer-based models have been adapted to graph inputs: Graphormer [26] incorporates graph structural encodings (e.g., spatial and centrality features) into the Transformer, achieving state-of-the-art performance on a variety of graph prediction tasks. This demonstrates the feasibility of large pretrained models on graph data, which could include networks of financial contracts or asset relationships. For tabular data, recent work introduces powerful foundation models as well. [27] present the Tabular Prior-Data Fitted Network (TabPFN), a transformer model pretrained on millions of synthetic tabular tasks; TabPFN “outperforms all previous methods” on diverse tabular benchmarks with up to 10k samples. As a generative model, TabPFN also supports fine-tuning, density estimation, and data generation, highlighting its versatility. The development of such graph and tabular foundation models implies that future DeFi risk models can integrate heterogeneous data: for example, combining graph neural networks on protocol networks with transformer encoders on protocol-level features. These advances provide a rich toolkit for building unified world

models of financial ecosystems that operate across structured graph data and tabular features.

3. Formalising Credit-Exposures on Decentralised Financial Networks

The hallmark of decentralised finance is that protocols lock and issue tokens that circulate across multiple platforms. When a lending protocol such as MakerDAO accepts wrapped ether (WETH) as collateral or a liquidity pool holds stablecoins minted by Circle or Tether, a web of implicit credit links arises: the value of one protocol’s liabilities depends on the solvency and liquidity of another. Traditional financial network models highlight that interconnectedness can diversify risk but also create channels for contagion; the amplification or dampening of shocks depends on network structure, leverage, and recovery rates [28]. Early systemic-risk research used clearing vectors to characterise interbank obligations and showed that fixed-point algorithms can compute consistent payment outcomes even in the presence of cyclical interdependence [29]. More recent work demonstrates that as networks become more complete, the stability of the system initially improves but may deteriorate beyond a critical threshold because dense interconnections propagate shocks [30]. These insights motivate a rigorous formalisation of credit-exposure in DeFi networks.

3.1. Credit Exposure

Let $X_G(q)$ denote the set of all tokens issued or generated by protocol q , and let $X_p(t)$ denote the set of tokens held (or locked) by protocol p at time t . Following [3], we say that protocol p has **credit exposure** to protocol q at time t if the tokens it holds include at least one that was issued by q :

$$X_p(t) \cap X_G(q) \neq \emptyset. \quad (1)$$

In other words, p holds a liability of q ; if the tokens issued by q lose value or become illiquid, then p faces a corresponding loss. This token-mediated dependence generalises the notion of counterparty credit risk in traditional finance to decentralised platforms, where exposures arise not from explicit bilateral contracts but from the possession of tokenised claims. The DeXposure dataset measures the dynamics of these relationships by reconstructing total value locked (TVL) time series for each protocol and tracking how the composition of tokens held by each protocol evolves over time.

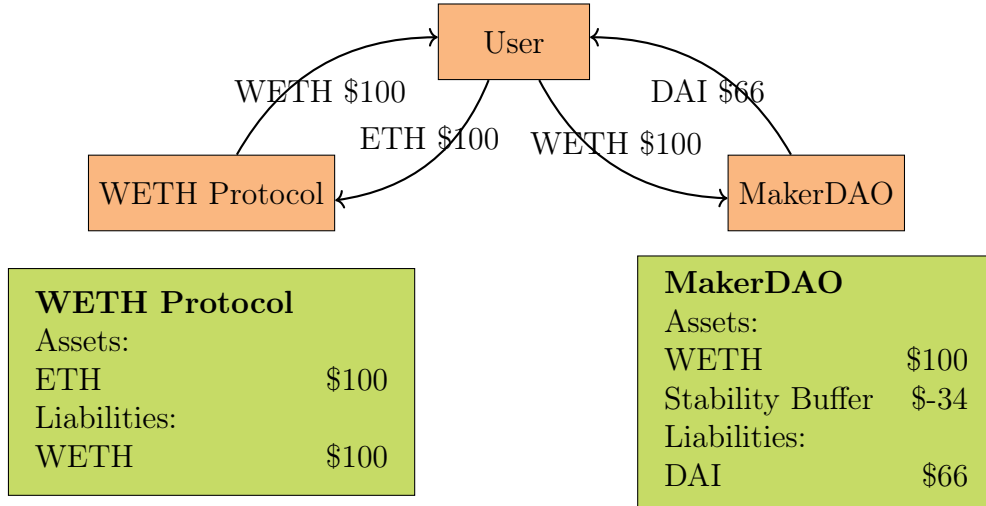


Figure 1: Credit Exposure Example: WETH Wrapping and Overcollateralized DAI Generation. User wraps ETH to WETH, then uses WETH as collateral to generate DAI from MakerDAO, creating credit exposure from MakerDAO to the WETH protocol.

Figure 1 (reproduced from [3]) illustrates how a simple user transaction creates a chain of exposures. A user wraps ether (ETH) into WETH via the WETH protocol and then deposits WETH into MakerDAO to generate the stablecoin DAI. In balance-sheet terms, the WETH protocol holds ETH as assets and issues WETH as liabilities, while MakerDAO holds WETH as collateral and issues DAI as liabilities. Because MakerDAO’s assets include WETH, there is credit exposure from MakerDAO to the WETH protocol; a failure or devaluation of WETH would erode MakerDAO’s collateral. This example shows that DeFi exposures are inherently multi-layered: a single wrapped token can create a cascade of dependencies across protocols and chains.

3.2. Mapping Tokens to Issuing Protocols

To operationalise (1), we must determine which protocol issues each token and thus identify the direction of exposure. Let $\mathcal{P}$ denote the set of all protocols and $\mathcal{C}$ the set of blockchains. For each token σ appearing in protocol p , a **mapping function** $M(\sigma)$ returns the protocol q that issues or manages σ . The DeXposure dataset constructs M using a fall-back procedure:

1. **Metadata lookup:** Tokens are directly matched to issuing protocols using metadata from DefiLlama when available.

2. **Manual mapping:** For tokens with high TVL that lack metadata links, researchers manually assign the issuing protocol based on documentation and expert knowledge.
3. **Text-vector similarity:** For remaining tokens, the text descriptions of tokens and protocols are vectorised via the term frequency-inverse document frequency (TF-IDF) representation [31]. Cosine similarity scores between token and protocol vectors determine the best match, and tokens are mapped to the protocol with the highest similarity score exceeding a threshold θ .
4. **Primary market tokens:** If none of the above methods yields a mapping, the token is treated as its own protocol (e.g., WETH).

This mapping ensures that token flows between protocols can be attributed to changes in credit exposure with respect to the correct issuing protocol, even when tokens are bridged across chains or wrapped by multiple platforms.

3.3. Network Representation and Value Flows

Credit exposures over a discrete time interval $\tau = t_2 - t_1$ are represented by a weighted directed graph $G_\tau = (P_\tau, E_\tau)$, where P_τ is the set of active protocols and E_τ is the set of directed edges. Each vertex $p \in P_\tau$ is assigned a **node weight** equal to the value of assets it holds at the end of the interval:

$$w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}, \quad (2)$$

where $v_{t_2, \sigma}$ is the USD value of token σ at time t_2 . Nodes with negligible weight (below a threshold θ) can be pruned for clarity. To define the **edge weight** from protocol p to protocol q , we first compute the value flow for each token σ :

$$F_{pq, \tau}^\sigma = \begin{cases} \max(0, -\Delta S_{p, \tau}^\sigma), & \text{if } \Delta S_{p, \tau}^\sigma < 0, \\ \max(0, \Delta S_{q, \tau}^\sigma), & \text{if } \Delta S_{q, \tau}^\sigma \geq 0, \end{cases} \quad (3)$$

where $\Delta S_{p, \tau}^\sigma = v_{p, t_2}^\sigma - v_{p, t_1}^\sigma$ is the change in the USD value of token σ held by protocol p over the interval. Equation (3) captures the intuition that if p decreases its holdings of σ and q increases theirs, then σ has flowed from p to q ; negative flows are reversed so that all flows are non-negative. The **edge weight** $w_\tau(e_{pq})$ is the sum of flows for all tokens that map to the issuing protocol q :

$$w_\tau(e_{pq}) = \sum_{\sigma \in X_{pq, \tau}} F_{pq, \tau}^\sigma, \quad (4)$$

where $X_{pq,\tau}$ denotes the set of tokens with issuing protocol q that move from p to q over τ . A positive edge weight indicates increasing exposure from p to q . For single-token protocols, the flow of that token equals the edge weight.

Algorithm 1 in [3] describes a procedure for constructing G_τ from raw token-protocol-chain mappings. In summary, one aggregates token holdings at the start and end of the interval, computes node weights, discards nodes below a threshold, and then computes edge weights by summing positive flows; negative flows are reversed to ensure all weights are non-negative. The resulting time-indexed sequence of graphs captures the evolving structure of credit dependencies in the DeFi ecosystem.

3.4. Discussion and links to systemic-risk theory

The above formalism bridges decentralised finance with established theories of financial networks. By modelling credit exposures as weighted directed graphs derived from token flows, it becomes possible to apply systemic-risk measures—such as centrality metrics, contagion indices, and stress tests, developed for interbank networks to DeFi. Classical models show that fixed-point algorithms can compute consistent clearing payments for interbank systems with cyclic obligations [29], and that network structures can either buffer or amplify shocks depending on leverage and liquidity [28]. [30] further demonstrate that more complete networks initially enhance stability by distributing losses across many counterparties, but beyond a critical density the same interconnections propagate shocks and increase systemic fragility. In the DeFi context, our graph formalisation reveals similar trade-offs: diversifying collateral across multiple protocols may reduce reliance on any single issuer, yet highly interconnected protocols could become conduits for contagion if a stablecoin or wrapped token suffers a de-peg or exploit.

This mathematical framework provides the foundation for the downstream tasks tackled by DeXposure-FM. Forecasting future node and edge weights, stress-testing scenarios under shocks to token values, and imputing missing TVL data all rely on accurate representations of credit exposure. Moreover, the network perspective aligns with macroprudential policy goals: regulators can monitor systemic importance and spillover risks by examining the temporal evolution of G_τ , and researchers can evaluate how proposed protocol designs or regulatory interventions would reshape the exposure network. As decentralised finance continues to grow and integrate with the broader financial system, rigorous formalisation and measurement of credit

exposure will be essential for safeguarding stability and fostering responsible innovation.

4. DeXposure-FM: Architecture, Data, and Training

This section introduces our DeXposure-FM, including its architecture, preparation of data, and the training process. The code is available at <https://github.com/EVIEHub/DeXposure-FM>.

4.1. DeXposure-FM Architecture

DeXposure-FM is a tri-modal foundation model that jointly represents (i) *protocol-level time series* (using time-series foundation model backbones [14, 21]), (e.g., TVL and its changes), (ii) *inter-protocol credit-exposure graphs* constructed from value flows, and (iii) *tabular protocol metadata* (using tabular foundation model backbones [27]). The model is designed to support three downstream objectives: (1) multi-step forecasting of edge weights and network statistics; (2) policy-relevant, shock-consistent scenario generation; and (3) imputation and denoising of missing or noisy histories.

4.1.1. Inputs and notation (aligned with the DeXposure construction)

Let $t \in \mathcal{T}$ denote discrete observation times and let $\tau = t_2 - t_1$ be a discrete interval between consecutive times. For each interval τ we construct a weighted directed graph snapshot

$$G_\tau = (\mathcal{P}_\tau, \mathcal{E}_\tau), \quad (5)$$

where $\mathcal{P}_\tau$ is the set of protocols and $\mathcal{E}_\tau$ is the set of directed exposure links. Node weights are given by $w_\tau(p)$ (the protocol’s end-of-interval stock) as in Eq. (2), and edge weights are given by $w_\tau(e_{pq})$ (the change in credit exposure from p to issuing protocol q over τ) as in Eq. (4), where token-level value flows $F_{pq,\tau}^\sigma$ are defined by Eq. (3). Credit exposure exists when $X_p(t) \cap X_G(q) \neq \emptyset$ (Eq. (1)).

For each protocol p we form three modality-specific inputs:

- **Time series window:** $\mathbf{x}_{p,\tau-L+1:\tau}^{\text{ts}}$ is a length- L history of protocol signals (e.g., $\{w_{\tau'}(p)\}_{\tau'}$, changes $\Delta S_{p,\tau'}^\sigma$, inflows/outflows, and sector aggregates).

- **Graph snapshot:** G_τ with node features including $w_\tau(p)$ and edge features including $w_\tau(e_{pq})$ (and optional attributes such as token class or sector pair).
- **Tabular metadata:** $\mathbf{x}_p^{\text{tab}}$ contains slowly varying or static descriptors (protocol category/sector, chain count, launch cohort, governance/bridge flags, etc.).

4.1.2. Tri-modal encoders with shared latent space

We parameterize DeXposure-FM with three pre-trained encoders (one per modality) and lightweight adapters that map each modality to a shared d -dimensional representation:

$$\mathbf{h}_p^{\text{tab}} = E_{\text{tab}}(\mathbf{x}_p^{\text{tab}}), \quad (6)$$

$$\mathbf{h}_{p,\tau}^{\text{ts}} = E_{\text{ts}}(\mathbf{x}_{p,\tau-L+1:\tau}^{\text{ts}}), \quad (7)$$

$$\tilde{\mathbf{h}}_p^{\text{tab}} = A_{\text{tab}}(\mathbf{h}_p^{\text{tab}}), \quad \tilde{\mathbf{h}}_{p,\tau}^{\text{ts}} = A_{\text{ts}}(\mathbf{h}_{p,\tau}^{\text{ts}}). \quad (8)$$

Adapters A_{tab} and A_{ts} (e.g., LayerNorm $\rightarrow$ bottleneck MLP) enable parameter-efficient fine-tuning while keeping the backbone encoders fixed or LoRA-tuned.

Gated fusion.. Because the informational content of time-series and tabular features varies across protocols and regimes, we fuse modality embeddings using a learned, elementwise gate:

$$\mathbf{g}_{p,\tau} = \sigma\left(W_g[\tilde{\mathbf{h}}_p^{\text{tab}}; \tilde{\mathbf{h}}_{p,\tau}^{\text{ts}}] + \mathbf{b}_g\right) \in (0, 1)^d, \quad (9)$$

$$\mathbf{z}_{p,\tau}^{(0)} = \mathbf{g}_{p,\tau} \odot \tilde{\mathbf{h}}_{p,\tau}^{\text{ts}} + (1 - \mathbf{g}_{p,\tau}) \odot \tilde{\mathbf{h}}_p^{\text{tab}}. \quad (10)$$

$\mathbf{z}_{p,\tau}^{(0)}$ serves as the initial node embedding for the graph encoder.

4.1.3. Graph encoder and spatiotemporal core

Given $(G_\tau, \{\mathbf{z}_{p,\tau}^{(0)}\})$, a pre-trained graph encoder (e.g., graph Transformer backbones [26, 15]) contextualizes each protocol within the contemporaneous exposure network:

$$\{\mathbf{z}_{p,\tau}\}_{p \in \mathcal{P}_\tau} = E_{\text{graph}}(G_\tau, \{\mathbf{z}_{p,\tau}^{(0)}\}_{p \in \mathcal{P}_\tau}). \quad (11)$$

We then model dynamics over a rolling window of graph snapshots using a temporal core (e.g., a temporal Transformer) applied node-wise:

$$\mathbf{s}_{p,\tau} = E_{\text{temp}}([\mathbf{z}_{p,\tau-L+1}, \dots, \mathbf{z}_{p,\tau}]). \quad (12)$$

The states $\mathbf{s}_{p,\tau}$ summarize a protocol’s recent temporal evolution *and* its network context.

4.1.4. Decoders and task heads

(1) *Multi-step forecasting..* For horizon $h \in \{1, \dots, H\}$ we predict future node weights $\hat{w}_{\tau+h}(p)$ and edge weights $\hat{w}_{\tau+h}(e_{pq})$. To scale edge prediction to large graphs, we use a low-rank bilinear decoder with nonnegativity:

$$\mathbf{u}_{p,\tau+h} = W_u \mathbf{s}_{p,\tau+h}, \quad \mathbf{v}_{q,\tau+h} = W_v \mathbf{s}_{q,\tau+h}, \quad (13)$$

$$\hat{w}_{\tau+h}(e_{pq}) = \text{softplus}(\mathbf{u}_{p,\tau+h}^\top \mathbf{v}_{q,\tau+h}). \quad (14)$$

Network-level statistics (concentration, density, sector connectivity) are obtained either (i) by applying deterministic functionals to the predicted graph $\hat{G}_{\tau+h}$, or (ii) via an auxiliary pooled head $H_{\text{net}}(\cdot)$ on a global embedding $\mathbf{g}_\tau = \text{Pool}(\{\mathbf{s}_{p,\tau}\})$.

(2) *Scenario-conditioned stress paths..* We represent a stress scenario as a structured shock vector $\mathbf{c}$ (e.g., stablecoin de-peg magnitude, collateral repricing, exchange outage indicator). A scenario encoder E_{scen} produces a context embedding that modulates the temporal core via FiLM:

$$\mathbf{s}'_{p,\tau} = \gamma(\mathbf{c}) \odot \mathbf{s}_{p,\tau} + \beta(\mathbf{c}), \quad (15)$$

and we decode a shock-consistent path $\{\hat{G}_{\tau+1}, \dots, \hat{G}_{\tau+H}\}$. We additionally regularize outputs to respect economic constraints (e.g., nonnegativity of flows and bounded changes in exposure) to improve scenario plausibility.

(3) *Imputation and denoising..* We train DeXposure-FM with masked-modality objectives: randomly mask entries in $\mathbf{x}^{\text{ts}}$ and in edge weights $w_\tau(e_{pq})$, and reconstruct them from the remaining modalities and network context. We report both reconstruction quality (TVL-weighted errors) and downstream gains in forecasting and link prediction.

4.2. Training Data Preparation

For each discrete interval $\tau = t_2 - t_1$ we construct a weighted directed snapshot $G_\tau = (\mathcal{P}_\tau, \mathcal{E}_\tau)$ with node weights $w_\tau(p)$ and edge weights $w_\tau(e_{pq})$. We form rolling training examples indexed by an anchor time τ :

$$\underbrace{\{G_{\tau-L+1}, \dots, G_\tau\}}_{\text{context window}} \longrightarrow \underbrace{\{G_{\tau+1}, \dots, G_{\tau+H}\}}_{\text{forecast targets}}, \quad (16)$$

where L is the context length and H is the forecast horizon. For each protocol $p \in \mathcal{P}_\tau$ we also provide modality-specific inputs: a protocol-level time-series window $\mathbf{x}_{p,\tau-L+1:\tau}^{\text{ts}}$, tabular protocol metadata $\mathbf{x}_p^{\text{tab}}$, and the contemporaneous exposure graph G_τ .

Time-based rolling splits. We evaluate models using time-based rolling splits (walk-forward validation), ensuring that all validation and test targets occur strictly after the corresponding training window to avoid look-ahead bias.

Scalable edge sampling. Since $|\mathcal{E}_\tau|$ can be large, we train with edge sampling. For each anchor τ and horizon h , we draw a set of positive edges from $\mathcal{E}_{\tau+h}$ with a mixture distribution that upweights large exposures (proportional to $w_{\tau+h}(e_{pq})$) while retaining a uniform component to preserve coverage of smaller links. When training an auxiliary link-existence head, we additionally sample non-edges as negatives.

4.3. Training Pipeline

This section describes the training pipeline.

4.3.1. Training phases

We use a three-phase training schedule to exploit pre-trained backbones while preventing catastrophic forgetting.

Phase A: adapter warm start (frozen backbones). We initialize the modality backbones (time-series, graph, tabular) from pre-trained checkpoints and freeze them. We train only lightweight adapters, fusion modules, the temporal core, and task heads until validation loss stabilizes.

Phase B: domain pretraining on DeXposure. We then pretrain on the full DeXposure history (excluding the final holdout block) using a mixture of masked reconstruction and predictive losses (Section 4.3.2). Backbones may remain frozen, or be tuned via parameter-efficient methods (e.g., LoRA/adapters) with a smaller learning rate.

Phase C: task fine-tuning. Finally, we fine-tune on the benchmark tasks reported in the paper: (i) multi-step forecasting of edge and node weights and network statistics; (ii) scenario-conditioned stress paths; and (iii) imputation/denoising. We either fine-tune jointly with a weighted multi-task objective, or sequentially with early stopping if negative transfer is observed.

4.3.2. Loss functions

We train DeXposure-FM with a weighted multi-task objective that reflects heavy-tailed magnitudes and the economic importance of large exposures.

Edge forecasting loss. We predict future edge weights $\widehat{w}_{\tau+h}(e_{pq})$ for horizons $h = 1, \dots, H$. Because exposures are heavy-tailed, we train in log space:

$$y_{\tau+h}(e_{pq}) = \log(1 + w_{\tau+h}(e_{pq})), \quad \hat{y}_{\tau+h}(e_{pq}) = \log(1 + \widehat{w}_{\tau+h}(e_{pq})). \quad (17)$$

We use a robust regression loss (Huber / SmoothL1) with horizon weights ω_h :

$$\mathcal{L}_{\text{edge}} = \sum_{h=1}^H \omega_h \mathbb{E}_{(p,q) \in \mathcal{S}_{\tau+h}} [\alpha_{pq,\tau+h} \text{Huber}(\hat{y}_{\tau+h}(e_{pq}) - y_{\tau+h}(e_{pq}))], \quad (18)$$

where $\mathcal{S}_{\tau+h}$ is the sampled edge set at horizon h . We set $\omega_h = \lambda^{h-1}$ with $\lambda \in (0, 1]$ to mildly downweight longer horizons.

Economic-importance weighting. To prioritize economically material nodes and edges, we weight errors by

$$\alpha_{pq,\tau} = (\log(1 + w_{\tau}(p)))^{\eta} (\log(1 + w_{\tau}(e_{pq})))^{1-\eta}, \quad (19)$$

with $\eta \in [0, 1]$, and clip $\alpha_{pq,\tau}$ to avoid domination by a handful of extreme edges.

Optional hurdle formulation (link + weight). To better handle sparsity, we optionally decompose forecasting into edge existence $a_{\tau+h}(p, q) \in \{0, 1\}$ and a conditional weight. The existence head is trained with binary cross entropy:

$$\mathcal{L}_{\text{link}} = \sum_{h=1}^H \omega_h \mathbb{E}_{(p,q) \in \mathcal{S}_{\tau+h}} [\text{BCE}(\hat{a}_{\tau+h}(p, q), a_{\tau+h}(p, q))], \quad (20)$$

and $\mathcal{L}_{\text{edge}}$ is applied only on positive edges.

Node forecasting loss. We forecast node “stock” weights $\widehat{w}_{\tau+h}(p)$ (and/or related node signals). Using the same log transform,

$$\mathcal{L}_{\text{node}} = \sum_{h=1}^H \omega_h \mathbb{E}_{p \in \mathcal{P}_{\tau+h}} [\beta_{p,\tau+h} \text{Huber}(\log(1 + \widehat{w}_{\tau+h}(p)) - \log(1 + w_{\tau+h}(p)))], \quad (21)$$

with $\beta_{p,\tau} = \log(1 + w_{\tau}(p))$ (clipped) to emphasize large protocols.

Network statistics loss. Let $\mathbf{m}_{\tau+h}$ denote a vector of network-level statistics (e.g., density, concentration, and sector connectivity). We train a pooled head $H_{\text{net}}(\cdot)$ to predict these statistics and encourage consistency with the statistics implied by the predicted graph:

$$\mathcal{L}_{\text{stats}} = \sum_{h=1}^H \omega_h \left\| \hat{\mathbf{m}}_{\tau+h}^{\text{head}} - \mathbf{m}_{\tau+h} \right\|_2^2 + \lambda_{\text{impl}} \sum_{h=1}^H \omega_h \left\| \hat{\mathbf{m}}_{\tau+h}^{\text{head}} - \hat{\mathbf{m}}_{\tau+h}^{\text{impl}} \right\|_2^2, \quad (22)$$

where $\hat{\mathbf{m}}_{\tau+h}^{\text{impl}} = \Phi(\hat{G}_{\tau+h})$ is computed deterministically from predicted edges and weights.

Masked reconstruction for imputation and denoising. We pretrain and fine-tune with masked-modality objectives by masking a subset of time-series entries $\mathcal{M}_{\text{ts}}$ and edge weights $\mathcal{M}_{\text{edge}}$:

$$\mathcal{L}_{\text{impute,ts}} = \mathbb{E}_{(p,t) \in \mathcal{M}_{\text{ts}}} \left[\gamma_{p,t} \text{Huber}(\hat{x}_{p,t}^{\text{ts}} - x_{p,t}^{\text{ts}}) \right], \quad (23)$$

$$\mathcal{L}_{\text{impute,edge}} = \mathbb{E}_{(p,q,\tau) \in \mathcal{M}_{\text{edge}}} \left[\alpha_{pq,\tau} \text{Huber}(\hat{y}_{\tau}(e_{pq}) - y_{\tau}(e_{pq})) \right], \quad (24)$$

$$\mathcal{L}_{\text{impute}} = \mathcal{L}_{\text{impute,ts}} + \lambda_{\text{impute,edge}} \mathcal{L}_{\text{impute,edge}}. \quad (25)$$

We report both reconstruction quality (including TVL-weighted errors) and downstream performance gains in forecasting and link prediction.

Scenario-conditioned stress loss and plausibility regularization. For scenario analysis we condition the model on a structured shock vector $\mathbf{c}$ (e.g., stable-coin de-peg magnitude, collateral repricing, exchange outage indicator) and train conditional multi-step predictions. When historical stress windows are available, we supervise scenario paths with the same forecasting losses:

$$\mathcal{L}_{\text{scen}} = \mathcal{L}_{\text{edge}}(\mathbf{c}) + \lambda_{\text{node}} \mathcal{L}_{\text{node}}(\mathbf{c}) + \lambda_{\text{stats}} \mathcal{L}_{\text{stats}}(\mathbf{c}). \quad (26)$$

To improve scenario plausibility, we include a smoothness regularizer on predicted log-weights:

$$\mathcal{L}_{\text{smooth}} = \sum_{h=2}^H \mathbb{E}_{(p,q) \in \mathcal{S}_{\tau+h}} \left| \hat{y}_{\tau+h}(e_{pq}) - \hat{y}_{\tau+h-1}(e_{pq}) \right|. \quad (27)$$

Nonnegativity of predicted flows and exposures is enforced by the decoder (e.g., via `softplus`).

Overall objective. The overall training objective is a weighted sum:

$$\mathcal{L} = \lambda_{\text{edge}}\mathcal{L}_{\text{edge}} + \lambda_{\text{link}}\mathcal{L}_{\text{link}} + \lambda_{\text{node}}\mathcal{L}_{\text{node}} + \lambda_{\text{stats}}\mathcal{L}_{\text{stats}} + \lambda_{\text{impute}}\mathcal{L}_{\text{impute}} + \lambda_{\text{scen}}\mathcal{L}_{\text{scen}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}}. \quad (28)$$

We tune $\{\lambda.\}$ on rolling validation windows; a robust starting choice is $\lambda_{\text{edge}} = 1$, $\lambda_{\text{node}} = 0.5$, $\lambda_{\text{stats}} = 0.2$, $\lambda_{\text{impute}} = 0.5$, $\lambda_{\text{scen}} = 0.5$, and $\lambda_{\text{smooth}} \in [0.01, 0.05]$.

4.3.3. Optimizer and learning-rate schedule

We optimize with AdamW and optionally apply sharpness-aware minimization (SAM) for improved generalization under regime shifts.

Base optimizer (AdamW). We use AdamW with $\beta_1 = 0.9$, $\beta_2 \in \{0.95, 0.999\}$, and weight decay (typically 10^{-2}) applied to non-normalization weights.

SAM wrapper. When using SAM, we wrap AdamW with neighborhood size $\rho \in [0.02, 0.1]$ (we use $\rho = 0.05$ unless otherwise stated). SAM performs two forward-backward passes per step and empirically improves robustness to distribution shift.

Parameter groups. We use two learning-rate groups: (i) newly introduced parameters (adapters, fusion, temporal core, heads) with learning rate $\text{lr}_{\text{new}} \approx 3 \times 10^{-4}$; and (ii) parameter-efficient backbone updates (e.g., LoRA/adapters inside pre-trained encoders) with a smaller learning rate $\text{lr}_{\text{backbone}} \in [5 \times 10^{-5}, 10^{-4}]$.

Schedule and stabilization. We apply linear warmup over the first 5–10% of training steps followed by cosine decay. We use gradient clipping (e.g., $\|\nabla\|_2 \leq 1$), mixed precision (bf16/fp16), and early stopping based on rolling-window validation loss.

4.4. Training Results

5. Empirical Evaluations

We evaluate DeXposure-FM as a foundation model for decentralized credit-exposure networks along three complementary axes: (i) *multi-step forecasting* of edge-level exposures and network-level statistics, (ii) *policy-relevant scenario analysis* that generates shock-consistent paths of losses and contagion under major stress events, and (iii) *imputation and denoising* of missing/noisy exposures and TVL histories, including downstream gains on forecasting and link-prediction.

Hyperparameter	Value / tuning range
Time step	Dataset interval τ (daily in DeXposure)
Context length L	32 (grid: {16, 32, 64})
Forecast horizon H	8 (grid: {4, 8, 12})
Targets	$\log(1 + w_\tau(p))$ and $\log(1 + w_\tau(e_{pq}))$
Robust loss	Huber / SmoothL1; $\delta = 1.0$ (0.5–2.0)
Horizon discount	$\omega_h = \lambda^{h-1}$ with $\lambda = 0.9$ (0.8–1.0)
Masking (pretraining)	TS: 15%; edges: 15% (10–30%)
Edge sampling	$\approx 5 \times 10^4$ positives per τ ; neg:pos = 3 : 1 (1–5)
Optimizer	AdamW; $(\beta_1, \beta_2) = (0.9, 0.95)$; wd = 10^{-2}
Learning rates	new params: 3×10^{-4} ; backbone/LoRA: 10^{-4} (5×10^{-5} – 10^{-4})
LR schedule	5% warmup + cosine decay
SAM (optional)	$\rho = 0.05$ (0.02–0.10)
Regularization	dropout = 0.1; grad clip $\ \nabla\ _2 \leq 1.0$; bf16
Early stopping	patience = 5 rolling validations
Loss weights	$\lambda_{\text{edge}} = 1, \lambda_{\text{link}} = 0.5, \lambda_{\text{node}} = 0.5, \lambda_{\text{stats}} = 0.2, \lambda_{\text{impute}} = 0.5, \lambda_{\text{scen}} = 0.5, \lambda_{\text{smooth}} = 0.02$

Table 1: Training hyperparameters (defaults and tuning ranges). All tuning is performed using time-based rolling validation.

5.1. Rolling-based Training and Evaluation

Let $\mathbf{E}_t \in \mathbb{R}_+^{N \times N}$ denote the directed exposure matrix at time t , where $(\mathbf{E}_t)_{ij}$ is the exposure of i to j , and let $\mathbf{x}_t$ collect node- and protocol-level covariates (e.g., TVL, yields, sector tags). We adopt a *time-based rolling-origin* evaluation scheme (walk-forward validation): for each forecasting origin τ we train on $\{1, \dots, \tau\}$ and evaluate on $\{\tau + 1, \dots, \tau + H\}$, rolling τ forward on a fixed stride. This aligns with standard out-of-sample forecast evaluation in time series [32, 33]. We report results under both (a) *expanding windows* and (b) *fixed-length sliding windows* to assess robustness to regime shifts.

For each metric we report the mean and dispersion across origins (e.g., mean $\pm$ sd). When comparing two forecasters, we assess statistical significance

using loss-differential tests following [34] with horizon-specific losses.

5.2. Task I: Multi-step Forecasting

Targets.. We evaluate forecasts at horizons $h \in \{1, 3, 7, 14\}$ (or matched to the dataset frequency) for:

1. **Edge-level exposures:** $\hat{\mathbf{E}}_{t+h}$.
2. **Node-level aggregates:** out-/in-exposure vectors $\mathbf{e}_t^{\text{out}} = \mathbf{E}_t \mathbf{1}$ and $\mathbf{e}_t^{\text{in}} = \mathbf{E}_t^\top \mathbf{1}$.
3. **Network-level statistics:** density, weighted density, concentration (e.g., exposure HHI), and sector connectivity (e.g., sector-by-sector exposure blocks).

Metrics (point forecasts).. We report MAE, RMSE, and (where well-defined) MAPE. To reflect economic materiality, we additionally report TVL- or exposure-weighted errors:

$$\text{WMAE} = \frac{\sum_{i,j} w_{ij,t} |(\hat{\mathbf{E}}_{t+h})_{ij} - (\mathbf{E}_{t+h})_{ij}|}{\sum_{i,j} w_{ij,t}}, \quad w_{ij,t} \propto \text{TVL}_{i,t} \text{ or } (\mathbf{E}_t)_{ij}.$$

We also report tail-focused metrics (e.g., top- k edges by realized exposure) and directional accuracy (sign of change) for node-level aggregates.

Distributional and structural fidelity.. Since exposures are heavy-tailed and network structure is central, we report (i) rank correlations of edge weights, (ii) errors on network statistics (e.g., HHI, density), and (iii) distributional distances between predicted and realized exposure-weight distributions (e.g., Wasserstein distance on $\log(1 + E)$). For sector connectivity we report block-wise errors and topological stability of the sector graph.

Baselines.. We compare against a broad set of classical and modern baselines:

- **Naïve** (persistence / random-walk) forecasts.
- **ARIMA**-family univariate forecasts [35].
- **VAR** / **SVAR** baselines on aggregated series and low-dimensional summaries [36, 37].
- **Dynamic factor** / **diffusion-index** forecasting using principal components [38].

- **Low-rank tensor/matrix forecasting** (dynamic matrix factorization; nuclear-norm regularization) [39].
- **Temporal graph neural networks** for dynamic graphs, including TGAT, TGN and DySAT variants [40, 41, 42].

All baselines use the same rolling-origin splits and are tuned on a validation segment inside each training window (or using nested rolling evaluation when feasible).

5.3. Task II: Policy-relevant Scenario Analysis

Scenario design.. We evaluate the model’s ability to generate *shock-consistent* counterfactual trajectories under targeted stress events:

1. **Stablecoin de-peg**: impose an exogenous price/peg shock and liquidity outflow pattern; propagate to protocols with stablecoin collateral and LP exposure.
2. **Exchange failure / large entity default**: impose a default or withdrawal shock on a major hub; model contagion through direct exposures and shared liquidity.

Scenarios are implemented as interventions on observed covariates (e.g., TVL, prices, volumes) and/or constraints on subsets of edges and node states, holding other exogenous series fixed.

Evaluation criteria.. Because scenarios are counterfactual, we evaluate along three dimensions:

- **Historical backtesting (event studies)**: for stress events observed in the sample, condition on pre-event history and compare generated trajectories to realized outcomes (losses, TVL drawdowns, exposure rewiring).
- **Internal consistency**: non-negativity, no-arbitrage style constraints where applicable, conservation rules for accounting identities, and monotone relations under larger shocks.
- **Contagion realism**: timing and localization of stress (which sectors and hubs are hit first), severity (peak-to-trough drawdown, cumulative losses), and breadth (fraction of nodes affected).

We summarize outcomes with interpretable indicators (e.g., peak drawdown of total TVL, time-to-peak-loss, fraction of protocols experiencing $> x\%$ drawdown) and compare to simplified contagion baselines inspired by systemic risk literature [30, 28].

5.4. Task III: Imputation and Denoising

Missingness and corruption regimes. We evaluate robustness under multiple regimes: (i) MCAR masking (random entries), (ii) block missingness (contiguous time gaps), and (iii) MNAR-like masking correlated with liquidity or stress (harder, policy-relevant). For denoising, we add controlled corruption (spikes, level shifts, and multiplicative noise) to exposures/TVL.

Reconstruction metrics. We compute MAE/RMSE on held-out entries and TVL-weighted variants (to prioritize economically material errors). We also report calibration diagnostics on reconstructed distributions (e.g., matching of heavy-tail behavior).

Downstream utility. To avoid optimizing solely for reconstruction, we measure downstream gains: (i) forecasting performance using imputed inputs, and (ii) link prediction quality (AUC, average precision) on edges whose existence/weight crosses economically meaningful thresholds. We report the *delta* versus baselines, isolating the effect of imputation quality on downstream tasks.

Imputation baselines. We compare against forward-fill / interpolation, KNN-style imputation, and low-rank matrix completion (Soft-Impute / nuclear norm) [39] as well as temporal GNN-based imputers [41, 40].

5.5. Ablations, Sensitivity, and Scalability

Ablations. We report ablations that isolate model components and modalities: (i) time-series only vs. time-series+graph, (ii) removing tabular covariates (TVL/sector tags), (iii) removing cross-asset conditioning, (iv) varying context length and prediction head, and (v) masking strategies during pre-training.

Generalization checks. We evaluate transfer across sub-networks (e.g., sectors, protocol families) and across regimes (stable vs. stress periods) by stratifying rolling windows and reporting performance dispersion.

Efficiency. We report training and inference cost (wall-clock, memory) as a function of N , number of edges, and sequence length, and we include throughput under batched inference for scenario simulation.

6. Financial Economics Insights

7. Conclusions

Acknowledgements

References

- [1] F. Schär, Decentralized finance: On blockchain- and smart contract-based financial markets, Federal Reserve Bank of St. Louis Review 103 (2) (2021) 153–174. doi:10.20955/r.103.153-74.
- [2] K. Gogol, C. Killer, M. Schlosser, T. Bocek, B. Stiller, C. Tessone, Sok: Decentralized finance (defi) – fundamentals, taxonomy and risks, <https://arxiv.org/abs/2404.11281> (2024). arXiv:2404.11281, doi:10.48550/arXiv.2404.11281.
- [3] W. Wu, K. Qian, A. Lui, C. Jack, Y. Wu, P. McBurney, F. He, B. Zhang, Dexposure: A dataset and benchmarks for inter-protocol credit exposure in decentralized financial networks, arXiv preprint arXiv:2511.22314 (2025). doi:10.48550/arXiv.2511.22314.
- [4] J. Bertomeu, X. Martin, I. Sall, Measuring defi risk, Finance Research Letters 63 (2024) 105321. doi:10.1016/j.frl.2024.105321.
- [5] S. Aufiero, S. Bartolucci, F. Caccioli, P. Vivo, Mapping microscopic and systemic risks in tradfi and defi: a literature review, arXiv preprint arXiv:2508.12007 (2025). doi:10.48550/arXiv.2508.12007.
- [6] Financial Stability Board, The financial stability risks of decentralised finance, Tech. rep., Financial Stability Board, accessed 27 December 2025 (2023).
URL <https://www.fsb.org/2023/02/the-financial-stability-risks-of-decentralise>
- [7] International Organization of Securities Commissions, Final report with policy recommendations for decentralized finance (defi), Tech. Rep. FR/14/2023, International Organization of Securities Commissions,

- accessed 27 December 2025 (Dec. 2023).
 URL <https://www.iosco.org/library/pubdocs/pdf/IOSCOPD754.pdf>
- [8] C. Duley, L. Gambacorta, R. Garratt, P. K. Wilkens, The oracle problem and the future of defi, BIS Bulletin 76, Bank for International Settlements, accessed 27 December 2025 (Sep. 2023).
 URL <https://www.bis.org/publ/bisbull76.pdf>
- [9] A. Sasi-Brodesky, I. K. Nassr, Defi liquidations: Volatility and liquidity, OECD Working Papers on Finance, Insurance and Private Pensions 48, OECD Publishing (Jul. 2023). doi:10.1787/0524faaf-en.
 URL https://www.oecd.org/en/publications/defi-liquidations_0524faaf-en.html
- [10] P. Tian, Y. Zhu, Liquidation mechanisms and price impacts in defi, Staff Working Paper 2025-12, Bank of Canada (Mar. 2025). doi:10.34989/swp-2025-12.
 URL <https://www.bankofcanada.ca/wp-content/uploads/2025/03/swp2025-12.pdf>
- [11] A. Kosse, M. Glowka, I. Mattei, T. Rice, Will the real stablecoin please stand up?, BIS Papers 141, Bank for International Settlements, accessed 27 December 2025 (Nov. 2023).
 URL <https://www.bis.org/publ/bppdf/bispap141.pdf>
- [12] K. Anadu, P. D. Azar, M. Cipriani, T. M. Eisenbach, C. Huang, M. Landoni, G. La Spada, M. Macchiavelli, A. Malfroy-Camine, J. C. Wang, Runs and flights to safety: Are stablecoins the new money market funds?, Staff Report 1073, Federal Reserve Bank of New York, revised April 2024; <https://doi.org/10.59576/sr.1073> (Sep. 2023). doi:10.59576/sr.1073.
 URL https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr1073.pdf
- [13] European Systemic Risk Board, Crypto-assets and decentralised finance: Report on stablecoins, crypto-investment products and multi-function groups, Tech. rep., European Systemic Risk Board, accessed 27 December 2025 (2025).

URL https://www.esrb.europa.eu/pub/pdf/reports/esrb.report202510_cryptoassets.en.pdf

- [14] Y. Liang, H. Wen, Y. Nie, Y. Jiang, M. Jin, D. Song, S. Pan, Q. Wen, Foundation models for time series analysis: A tutorial and survey, arXiv preprint arXiv:2403.14735KDD 2024 tutorial and survey; <https://arxiv.org/abs/2403.14735> (2024).
- [15] D. Ereemeev, G. Bazhenov, O. Platonov, A. Babenko, L. Prokhorenkova, Turning tabular foundation models into graph foundation models, arXiv preprint arXiv:2508.20906<https://arxiv.org/abs/2508.20906> (2025).
- [16] D. Oikonomou, N. Loizou, Sharpness-aware minimization: General analysis and improved rates, in: International Conference on Learning Representations (ICLR), 2025, openReview; <https://openreview.net/forum?id=...>
- [17] N. Naifar, Mapping systemic tail risk in crypto markets: Defi, stablecoins, and infrastructure tokens, Journal of Risk and Financial Management 18 (6) (2025) 329, <https://www.mdpi.com/1911-8074/18/6/329>. doi:10.3390/jrfm18060329.
- [18] S. Aufero, S. Bartolucci, F. Caccioli, P. Vivo, Mapping microscopic and systemic risks in tradfi and defi: a literature review, arXiv preprint arXiv:2508.12007<https://arxiv.org/abs/2508.12007> (2025).
- [19] J. Ding, Y. Zhang, Y. Shang, Y. Zhang, Z. Zong, J. Feng, Y. Yuan, H. Su, N. Li, N. Sukiennik, F. Xu, Y. Li, Understanding world or predicting future? a comprehensive survey of world models, arXiv preprint arXiv:2411.14499 (2024). doi:10.48550/arXiv.2411.14499.
- [20] A. Carriero, D. Pettenuzzo, S. Shekhar, Macroeconomic forecasting with large language models, arXiv preprint arXiv:2407.00890 (2025). doi:10.48550/arXiv.2407.00890.
- [21] A. Das, W. Kong, R. Sen, Y. Zhou, A decoder-only foundation model for time-series forecasting, arXiv preprint arXiv:2310.10688 (2023). doi:10.48550/arXiv.2310.10688.

- [22] Z. Zhu, H. Chen, Q. Qu, V. Chung, Fincast: A foundation model for financial time-series forecasting, arXiv preprint arXiv:2508.19609 (2025). doi:10.48550/arXiv.2508.19609.
- [23] M. Faw, R. Sen, Y. Zhou, A. Das, In-context fine-tuning for time-series foundation models, in: Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), 2025.
- [24] V. Balmaseda, M. Coronado, G. de Cadenas-Santiago, Predicting systemic risk in financial systems using deep graph learning, *Intelligent Systems with Applications* 19 (2023) 200240. doi:10.1016/j.iswa.2023.200240.
- [25] L. Gonon, T. Meyer-Brandis, N. Weber, Computing systemic risk measures with graph neural networks, arXiv preprint arXiv:2410.07222 (2024). doi:10.48550/arXiv.2410.07222.
- [26] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, T.-Y. Liu, Do transformers really perform bad for graph representation? a case study on graphormer, in: *Advances in Neural Information Processing Systems* (NeurIPS 2021), 2021.
- [27] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Korfer, S. B. Hoo, R. T. Schirrmeister, F. Hutter, Accurate predictions on small data with a tabular foundation model, *Nature* 637 (2025) 319–326. doi:10.1038/s41586-024-08328-6.
- [28] P. Glasserman, H. P. Young, Contagion in financial networks, *Journal of Economic Literature* 54 (3) (2016) 779–831. doi:10.1257/jel.20151228.
- [29] L. Eisenberg, T. H. Noe, Systemic risk in financial systems, *Management Science* 47 (2) (2001) 236–249. doi:10.1287/mnsc.47.2.236.9835.
- [30] D. Acemoglu, A. Ozdaglar, A. Tahbaz-Salehi, Systemic risk and stability in financial networks, Working Paper 18727, National Bureau of Economic Research (2013). doi:10.3386/w18727.
URL <https://www.nber.org/papers/w18727>

- [31] S. Qaiser, R. Ali, Text mining: Use of tf-idf to examine the relevance of words to documents, *International Journal of Computer Applications* 181 (1) (2018) 25–29. doi:10.5120/ijca2018917395.
- [32] L. J. Tashman, Out-of-sample tests of forecasting accuracy: an analysis and review, *International Journal of Forecasting* 16 (4) (2000) 437–450. doi:10.1016/S0169-2070(00)00065-0.
- [33] R. J. Hyndman, G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd Edition, OTexts, 2021.
URL <https://otexts.com/fpp3/>
- [34] F. X. Diebold, R. S. Mariano, Comparing predictive accuracy, *Journal of Business & Economic Statistics* 13 (3) (1995) 253–263. doi:10.1080/07350015.1995.10524599.
- [35] G. E. P. Box, G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Holden-Day, San Francisco, 1970.
- [36] C. A. Sims, Macroeconomics and reality, *Econometrica* 48 (1) (1980) 1–48. doi:10.2307/1912017.
- [37] H. Lütkepohl, *New Introduction to Multiple Time Series Analysis*, Springer, Berlin, 2005. doi:10.1007/978-3-540-27752-1.
- [38] J. H. Stock, M. W. Watson, Forecasting using principal components from a large number of predictors, *Journal of the American Statistical Association* 97 (460) (2002) 1167–1179. doi:10.1198/016214502388618960.
- [39] R. Mazumder, T. Hastie, R. Tibshirani, Spectral regularization algorithms for learning large incomplete matrices, *Journal of Machine Learning Research* 11 (2010) 2287–2322.
URL <http://jmlr.org/papers/v11/mazumder10a.html>
- [40] D. Xu, C. Ruan, E. Korpeoglu, S. Kumar, K. Achan, Inductive representation learning on temporal graphs, *arXiv preprint arXiv:2002.07962* (2020). doi:10.48550/arXiv.2002.07962.
- [41] E. Rossi, B. P. Chamberlain, F. Frasca, D. Eynard, F. Monti, M. Bronstein, Temporal graph networks for deep learning on dynamic graphs, *arXiv preprint arXiv:2006.10637* (2020). doi:10.48550/arXiv.2006.10637.

- [42] A. Sankar, Y. Wu, L. Gou, W. Zhang, H. Yang, Dynamic graph representation learning via self-attention networks, arXiv preprint arXiv:2005.07241 (2020). doi:10.48550/arXiv.2005.07241.